Major Project – 2
REPORT

Project Title:
Domain-Specific RAG Chatbot

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1.Introduction

The Domain-Specific Retrieval-Augmented Generation (RAG) Chatbot is an AI-based application designed to answer questions using information from uploaded PDF documents. It combines document retrieval with a Large Language Model (LLM) to generate relevant answers based on the content of the documents.

2. Objectives

• Upload and extract text from PDF documents.
• Retrieve relevant information based on user questions.
• Generate context-based answers using AI.
• Display source document and page information when available.
• Reduce the time required to search information manually.

3. Technologies Used

The project uses **Python** as the programming language and **Streamlit** for the user interface. PyPDF extracts text from PDF documents, LangChain Text Splitters divide the text into smaller chunks, Sentence Transformers generates vector embeddings, and FAISS performs similarity search. The Groq API is used to generate answers from the retrieved information.

4. Methodology

The user uploads PDF documents through the Streamlit application. The system extracts the text and divides it into smaller chunks. Sentence Transformers converts these chunks into numerical vector embeddings, which are indexed using FAISS. When the user asks a question, the system retrieves the most relevant text chunks and sends the retrieved context along with the question to the configured language model. The generated answer and available source details are then displayed to the user.

5. Results and Applications

The application provides an interactive way to ask questions about uploaded documents and retrieve relevant information. It can be useful for students, researchers, and professionals who need to search notes, study materials, reports, and other text-based PDF documents. Questions that cannot be answered from the retrieved document content are handled using a predefined fallback response.

6. Conclusion

The Domain-Specific RAG Chatbot demonstrates how retrieval-based search and generative AI can work together to make document-based information retrieval faster and easier. Future enhancements include OCR integration for scanned and handwritten PDFs, improved retrieval accuracy, and support for additional document formats.